Annals of Formalized Mathematics
Volume **0** (2026), 1–6
DOI: https://doi.org/10.46298/afm.?????

A Conditional Formal Reduction of Legendre's Conjecture via the Krafft-Turán Sieve

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Abstract. We present a fully formalized, conditional reduction of Legendre's Conjecture in the Lean 4 proof assistant. Legendre's Conjecture asserts the existence of at least one prime between consecutive perfect squares n^2 and $(n+1)^2$. By deploying a formulation of the Turán sieve intertwined with the Krafft geometric intervals, we establish a "zero-sorry" formal scaffold bounding the sieve capacity of large primes. While an unconditional proof requires establishing prime gaps of order $O(x^{1/2})$—a well-known open problem in analytic number theory—our formalized reduction cleanly separates the absolute, verified geometric properties from the unproven analytic density estimate. Consequently, we mathematically guarantee that Legendre's Conjecture holds conditionally upon this isolated analytic threshold.

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MSC 2020: 11N05, 11N35, 11N36, 03B35
Keywords: Legendre's Conjecture, Krafft intervals, Turán sieve, Prime gaps, Lean 4, Formalization

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Received: 00
Final form: 00
Accepted: 00

1 Introduction

Legendre's Conjecture is one of the oldest open problems in number theory, postulating that for every positive integer n, there exists at least one prime p such that $n^2 < p < (n+1)^2$. An unconditional proof of this conjecture remains out of reach, as it intrinsically demands controlling the gaps between consecutive primes strictly within $O(x^{1/2})$, whereas the best unconditional bounds currently available are of order $O(x^{0.525})$ [BHP01].

In this manuscript, we offer a *formal, conditional reduction* mechanized in the Lean 4 interactive theorem prover. Our approach translates the spatial bounds of Legendre's Conjecture into a systematic geometric framework via *Krafft* [Kra01; Por26] *sub-intervals*, integrating this geometry natively into a formalized Turán sieve. We successfully formalize a contradiction scaffold that definitively proves Legendre's Conjecture *conditional* upon a specific, isolated inequality representing the expected analytic density of sieve survivors.

2 Krafft Geometry and Legendre Targeting

The bedrock of our formalization is the explicit partitioning of the spatial domain into six distinct Krafft sub-intervals, denoted $B_n^{(k)}$ for $k \in \{1, \ldots, 6\}$. These sets partition the surrounding moduli space to tightly restrict the residue classes.

Our first primary formalized milestone is the *Legendre targeting property* (`legendre_targeting`).

***Theorem* 2.1** (Legendre Targeting Property)**.** *For any positive integer $n \geq 1$, Krafft coordinate index $k \in \{0, \ldots, 5\}$, and any integer $x \in B_n^{(k)}$, the prime candidates natively satisfy:*

$$(6n + k - 1)^2 < 6x - 1 \quad \textit{and} \quad 6x + 1 < (6n + k)^2$$

Proof. As formalized in Lean 4, this holds strictly via algebraic expansion. The boundaries of $B_n^{(k)}$ tightly constrain x such that the geometric mapping $6x \pm 1$ inevitably lands unconditionally within the gap between the squares of consecutive integers. Thus, any prime found at these targeted coordinates strictly satisfies Legendre's requirement. □

This combinatorial translation means that proving Legendre's Conjecture is reduced to formally verifying that for sufficiently large n, *every* Krafft interval independently generates at least one survivor x such that either $6x - 1$ or $6x + 1$ is prime.

3 The Cardinality Contradiction

To isolate surviving prime candidates, we formalize a combinatorial exhaustion assuming the negation of Legendre's Conjecture—a scenario we term the "Double Cover". Under a Double Cover, every geometric candidate in the target subset must be composite, meaning it is struck by at least one large sieving prime $p > 2n$.

By analyzing the difference between paired residue strikes, we formalize a rigid spatial vector connecting potential dual-strikes by the same prime p (`paired_strike_separation`). Because traversing this modular distance requires steps far exceeding the structural width of $B_n^{(k)}$, we formally derive the "Wasted Cover" limit.

***Lemma* 3.1** (Krafft Strike Separation)**.** *If a prime $p \in P_{large}$ strikes a positive candidate $6x + 1 \equiv 0 \pmod{p}$ and a distinct negative candidate $6y - 1 \equiv 0 \pmod{p}$ within the same interval $x, y \in B_n^{(k)}$, their spatial separation must rigorously be:*

$$x - y \equiv 2\alpha m \pmod{p}$$

where $p = 6m + \alpha$.

Proof. By constructing the translation vector strictly within $\mathbb{Z}/p\mathbb{Z}$, the modular requirement evaluates exactly to $2\alpha m$. Since $p > 2n$, this structural requirement forces an absolute geometric distance $|x - y| \geq (p - 1)/3$. However, the interval $B_n^{(k)}$ has maximum width bounded safely below this. A structural spatial contradiction guarantees that no such dual-strike can occur, legally formalizing that each large prime captures at most one candidate from the positive residue class and one from the negative class. □

This geometric ceiling on strike efficiency yields a hard upper bound on the maximum number of candidates the large primes can eliminate. If the analytic density of points surviving the initial small prime sieving strictly exceeds the combined covering capacity of the available large primes, the Double Cover assumption mathematically collapses, guaranteeing the survival of at least one prime (`legendre_conjecture_holds_conditional`).

***Theorem* 3.2** (Conditional Legendre's Conjecture)**.** *Assuming there exists a constant N_0 such that for all $n \geq N_0$, the analytic density of candidates surviving the initial small-prime sieve locally exceeds the covering summation bounds of the large sieve prime reciprocals, then Legendre's Conjecture firmly holds for all $n \geq N_0$.*

Proof. By sealing the Double Cover exhaustion inside Lean 4, we generate a final cardinality contradiction. The sum of the available large-prime strikes explicitly fails to cover the geometric target sets U^+ and U^-, enforcing that the pigeonhole principle formalized over these finite discrete spaces must output an un-struck non-composite coordinate. □

4 Conclusion

We have successfully constructed a fully verified, "zero-sorry" formal scaffold bridging structural sieve geometry with the constraints of analytic number theory. By sealing off the conditional analytic estimate into an explicit hypothesis, the Lean 4 formalization explicitly guarantees that if the analytic inequality is resolved, Legendre's Conjecture follows unyieldingly from the verified mechanics. This project underscores the utility of formal methods not merely in verifying completed proofs, but in structurally isolating the exact boundaries of open problems.

Acknowledgements

This work would not have been possible without the tireless assistance of the Aristotle framework and Gemini 3.1 Pro. The author wishes to express his gratitude for their unwavering dedication to parsing the intricate complexities of the formalization, illuminating structural insights, and maintaining a steadfast, collaborative dialogue. More than mere computational instruments, they served as indispensable collaborators whose tireless deductive logic helped chart the course through these formalized geometric spaces. Specifically, Gemini 3.1 Pro was instrumental in clarifying the theoretical ideas and generating precise prompt heuristics for the formal prover, while Aristotle served as the primary mechanical workhorse securing the dense derivations.

References

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Appendix: Dependencies between theorems

The following directed graphs map the logical dependencies between the central formalized results. To maintain structural clarity and avoid overwhelming edge density, basic definitions and foundational types are generally excluded from the hierarchy.

- **Blue-shaded rounded rectangles**: Formal proofs (Lemmas and Theorems).
- **Green-shaded rectangles**: Central definitions or structures.
- **Directed arrows** ($A \to B$): Logical dependence, signifying that proof A explicitly "uses" or "applies" result B to close a goal.

Figure 1: .